

M1 Conversion Core Result

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Purpose. This document reports the results of the conversion function and tests built according to the M1 design [D1], confirms whether the completion criteria are met, and asks for approval to close M1 and move on to the M2 design.

1 Summary

`convert.py` (40 lines) and `test_convert.py` (74 lines) were built as designed. **All five completion criteria were met.**

- **Accuracy.** All 9 defined values matched with zero error, and the maximum relative error over the 5 temperature reference points was $1.4\text{e-}15$, about 700 times smaller than the tolerance of $1\text{e-}12$.
- **Round trip.** For all 109 same-kind unit pairs, $25 \rightarrow B \rightarrow A$ returned to 25. The maximum relative error is $1.4\text{e-}16$.
- **Test strength.** Replacing the pound factor with the rounded NIST value (0.4535924) made the tests fail. The tests catch a factor error in the 7th digit.
- **Dependencies.** `convert.py` has 0 imports.

2 Goal and completion criteria

The goal of M1 was `convert(value, src, dst)`, callable without the command line, and its tests [D1]. Table 1 lists the completion criteria from Table 5 of the design document and the verdicts.

Table 1. Completion criteria and verdicts. [D1, D3]

Criterion	Target	Result	Verdict
Defined values	9 / 9 pass	9 / 9	met
Temperature reference points	5 / 5 pass	5 / 5	met
Round trip	109 / 109 pass	109 / 109	met
Errors and naming rules	4 / 4 pass	4 / 4	met
Dependencies	0 imports	0	met

3 Work done

The work followed the order in the design (tests first → tables and function → full run). The statuses in the boxes below are adopted / option / rejected / deferred. No attempt failed and was discarded. The failure on the first test run was intended: `convert.py` did not exist yet [D3].

3.1 Conversion code (`convert.py`)

① Factor table and temperature formula table (M1.1) — adopted

Reason. With n units, writing only n relations to the base unit is enough to convert every pair [D1].

Method. `LENGTH` (8 units), `MASS` (6) and `TEMP` (3, each with a pair of to-kelvin/from-kelvin formulas) were written exactly as in design Table 2. One more dictionary, `KINDS`, maps kind names to tables, so the same dictionary serves both to find which kind a unit belongs to and to build the unit list for the “unknown unit” error.

Result. The 9 defined-value tests passed with zero error. $100 \text{ km} \rightarrow \text{mi} = 62.13711922373339$.

② 0 K check before the same-unit return (M1.2) — adopted

Reason. The design specifies that a conversion between identical units returns the value unchanged (design Section 3.4). If this return came first, `-300 c → c` would skip the 0 K check and return a temperature that cannot exist.

Method. The order was set to “look up units → check kind → (if temperature) 0 K check → return unchanged if same unit → compute”.

Result. `convert(-300, "c", "c")` raised the error “-300 c is below absolute zero”. `-273.15 c` (exactly 0 K) was accepted.

3.2 Tests (test_convert.py)

③ Comparison with tolerance (M1.3) — adopted

Reason. During design, `100 °C → °F` computed to `211.99999999999994`, so an exact equality comparison would fail even for correct code [D1].

Method. All comparisons used `math.isclose(rel_tol=1e-12, abs_tol=1e-12)` [R1]. The expected values for the defined-value tests were copied separately from proposal Table 3, not from `convert.py` [D2].

Result. All 127 cases checked by the 7 test methods (9 defined values + 5 reference points + 109 round trips + 4 errors and names) passed. The run time is 0.001 s.

④ Checking the tests with wrong factors (M1.4) — adopted

Reason. A passing test is useless if it does not fail when a factor is slightly wrong. The round-trip test in particular passes even with a wrong factor, because the forward and backward errors cancel.

Method. The files were left unchanged; one factor was changed only at run time and the full test suite was run. (a) The pound was set to the rounded NIST table value `0.4535924` [R2]; (b) the foot was set to the US survey foot `0.3048006` [R2].

Result. In both cases one defined-value test failed and the round-trip tests passed, as expected. Factor errors are caught by the defined-value tests.

4 Results

Figure 1 compares the maximum relative error of each test group with the tolerance. The values differ by powers of ten, so the vertical axis is logarithmic. Table 2 lists all

measurements.

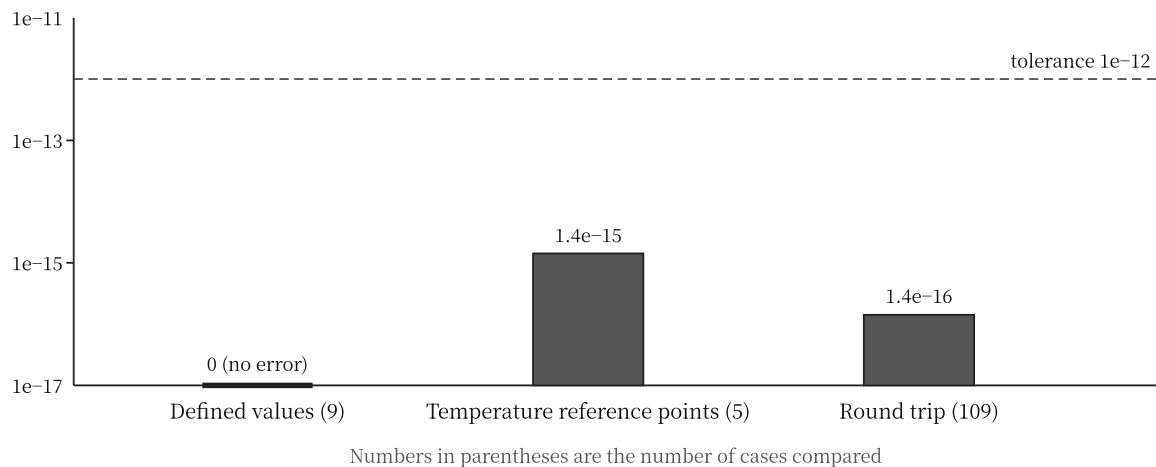


Figure 1. Maximum relative error by test group (log scale). All three groups lie roughly three orders of magnitude below the dashed tolerance line. Defined values have exactly zero error, so they have no bar.

Table 2. Measurements. Python 3.11.16, measured 2026-10-09. [D3]

Item	Measurement	Value
Defined-value error	Maximum relative error over 9 values	0
Temperature reference point error	Maximum: $-40\text{ }^{\circ}\text{C} \rightarrow ^{\circ}\text{F} = -40.000000000000006$	1.4e-15
Round-trip error	Maximum over 109 pairs; 2 pairs nonzero	1.4e-16
Factor error detection	Tests failed with each of 2 wrong factors	1 each
Test run time	<code>python -m unittest</code> , 7 methods	0.001 s
Time per call	Mean of 100,000 calls, temperature / length	0.62 / 0.49 μs
Code size	<code>convert.py</code> / <code>test_convert.py</code>	40 / 74 lines

5 Discussion

Writing the defined values directly leaves almost no error for length and weight. The 9 defined values had zero error, and only 2 of the 109 round-trip pairs had nonzero error. The remaining error comes from temperature, because its formulas mix addition and multiplication; even so, it is about 700 times smaller than the tolerance. Design Table 6 listed the same case as 1.5e-15, but an exact recalculation gives 1.4e-15. This does not affect the verdict.

Round-trip tests alone do not catch factor errors. A wrong factor cancels out between the forward and return trips. What protects the factors is the defined-value tests with separately

written expected values, as confirmed in Section 3.2 ④.

There is one difference from the design. Error messages show unit names in lowercase (entering `xx` displays `'xx'`). This follows from the case-insensitive rule. It can be changed in M2 if showing the input as typed is preferred.

The evidence has limits. The test values are only 25, the defined values and the reference points. Very large values (1e300) and very small values were not checked (not verified). The source for the pound definition is a secondary source [D2].

6 Next steps

Once M1 is approved, work moves on to the M2 design. Table 3 lists the work M2 takes over.

Table 3. Next steps.

Task	Content	Needed
M2 design	Take a value and two units with <code>argparse</code> , print to 10 significant digits, and catch <code>ValueError</code> to exit with exit code 2	Approval
Non-numeric value	The one of the proposal's 4 error types deferred from M1	Included in M2 design
Unit spelling in error messages	Show in lowercase or as typed	Developer decision (default: lowercase, as now)

References

[R1] Python Software Foundation. `math.isclose`.

<https://docs.python.org/3/library/math.html#math.isclose>

[R2] NIST. NIST Guide to the SI, Appendix B.8 / B.9 (SP 811). <https://www.nist.gov/pml/special-publication-811> (accessed 2026-10-09)

[D1] M1 Conversion Core Design (docs/reports/unit-converter-cli/M1-design.pdf), approved 2026-10-09.

[D2] Unit Conversion CLI Proposal (docs/reports/unit-converter-cli/00-proposal.pdf), approved 2026-10-09.

[D3] `convert.py`, `test_convert.py` and measurement run logs, Python 3.11.16, 2026-10-09 (uncommitted).